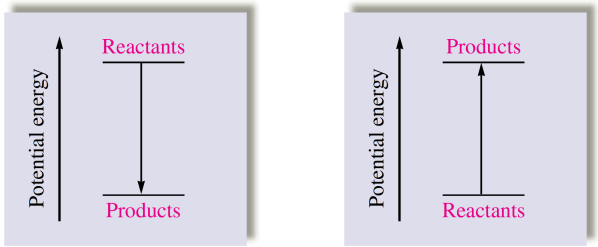
**Name:** _____ **Favorite Temperature:** _____

*Thermodynamics Take Home Quiz (Time Yourself and Complete the whole thing in 1 Hour 35 Minutes)*

**Multiple Choice (MCQ)** *1 pt each*

1. Which plot represents an exothermic reaction?




**a.** **b.**

(a) Plot a

(b) Plot b

(c) Both plot a and plot b

(d) Neither plot a nor plot b

2. Calculate $\Delta E$ for a system that absorbs 82 kJ of heat and does 47 kJ of work.

(a) 129 kJ

(b) 35 kJ

(c) −35 kJ

(d) −129 kJ

3. One hundred joules of work are required to compress a gas. At the same time, the gas releases 23 J of heat. Calculate the internal energy change, $\Delta E$.

(a) +123 J

(b) −123 J

(c) −77 J

(d) +77 J

4. Consider the following reaction:

$$2\,H_2(g) + O_2(g) \longrightarrow 2\,H_2O(l) \quad \Delta H = -572 \text{ kJ}$$

How much heat is evolved for the production of 1.00 mol of $H_2O(l)$?

(a) 572 kJ

(b) 1144 kJ

(c) 286 kJ

(d) −286 kJ

5. The specific heat capacity of silver is 0.24 J/°C · g. Calculate the energy required to raise the temperature of 150.0 g of Ag from 273 K to 298 K.

(a) 900 J

(b) 9828 J

(c) 10.7 kJ

(d) 36 J

6. In a coffee-cup calorimeter, 50.0 mL of 0.100 M $AgNO_3$ and 50.0 mL of 0.100 M HCl are mixed. The temperature rises from 22.60°C to 23.40°C. Calculate the heat (kJ/mol) for:

$$Ag^+(aq) + Cl^-(aq) \longrightarrow AgCl(s)$$

(Assume 100.0 g solution, $C_p = 4.18$ J/°C · g).

(a) −66 kJ/mol

(b) +66 kJ/mol

(c) −334 kJ/mol

(d) −0.334 kJ/mol

7. Calculate $\Delta H$ for the reaction:

$$2\,C(s) + O_2(g) \longrightarrow 2\,CO(g)$$

Given: $C(s) + O_2(g) \longrightarrow CO_2(g)$ $\Delta H = -393.7$ kJ/mol and $CO(g) + \frac{1}{2}\,O_2(g) \longrightarrow CO_2(g)$ $\Delta H = -283.3$ kJ/mol.

(a) $-110.4$ kJ

(b) $-220.8$ kJ

(c) $-677.0$ kJ

(d) $+110.4$ kJ

8. Which of the following reactions has $\Delta H^\circ$ values equal to $\Delta H_f^\circ$ for the product?

(a) $Na(s) + \frac{1}{2}\,Cl_2(g) \longrightarrow NaCl(s)$

(b) $Na(g) + \frac{1}{2}\,Cl_2(g) \longrightarrow NaCl(s)$

(c) $2\,Na(s) + Cl_2(g) \longrightarrow 2\,NaCl(s)$

(d) $Na^+(aq) + Cl^-(aq) \longrightarrow NaCl(s)$

9. For the process $H_2O(l) \longrightarrow H_2O(g)$ at 298 K and 1.0 atm, $\Delta H$ is more positive than $\Delta E$ by 2.5 kJ/mol. What does this 2.5 kJ/mol represent?

(a) Heat of fusion

(b) Work done by the system to push back the atmosphere

(c) Activation energy

(d) Entropy change

10. Calculate the enthalpy change for the reaction:

$$CH_4(g) + 2\,O_2(g) \longrightarrow CO_2(g) + 2\,H_2O(l)$$
$$\Delta H = -891 \text{ kJ}$$

When 1.00 g of methane is burned in excess oxygen.

(a) $-891$ kJ

(b) $-55.7$ kJ

(c) $-14.3$ kJ

(d) $+55.7$ kJ

11. Predict the sign of $\Delta S_{surr}$ for the process: $H_2O(l) \longrightarrow H_2O(g)$.

(a) Positive

(b) Negative

(c) Zero

(d) Depends on the temperature

12. Calculate $\Delta S_{surr}$ for the following reaction at 25°C and 1 atm:

$$C_3H_8(g) + 5\,O_2(g) \longrightarrow 3\,CO_2(g) + 4\,H_2O(l)$$
$$\Delta H^\circ = -2221 \text{ kJ}$$

(a) 7.45 J/K

(b) 7450 J/K

(c) $-7450$ J/K

(d) 89.3 J/K

13. For mercury, the enthalpy of vaporization is 58.51 kJ/mol and the entropy of vaporization is 92.92 J/K · mol. What is the normal boiling point of mercury?

(a) 0.630 K

(b) 630 K

(c) 357 K

(d) 630°C

14. Predict the sign of $\Delta S^\circ$ for the following reaction:

$$2\,H_2(g) + O_2(g) \longrightarrow 2\,H_2O(l)$$

(a) Positive

(b) Negative

(c) Zero

(d) Cannot be predicted

15. Which substance has the greater value of $S^\circ$?

(a) $C_2H_5OH(l)$

(b) $C_2H_5OH(g)$

(c) Both are equal

(d) $C_2H_5OH(s)$

16. Calculate $\Delta G^\circ$ for the reaction:

$$2\,NO_2(g) \rightleftharpoons N_2O_4(g)$$

Given $\Delta H^\circ = -58.03$ kJ and $\Delta S^\circ = -176.6$ J/K at 298 K.

(a) $-5.40$ kJ

(b) $-110.6$ kJ

(c) $+5.40$ kJ

(d) $-58.2$ kJ

17. At what temperature is $\Delta G^\circ = 0$ for the reaction $2\,NO_2(g) \rightleftharpoons N_2O_4(g)$? (Assume $\Delta H^\circ = -58.03$ kJ and $\Delta S^\circ = -176.6$ J/K).

(a) 0.328 K

(b) 328.6 K

(c) 298 K

(d) 3.04 K

18. Using the equation $\Delta G^\circ = -RT\ln(K)$, calculate $\Delta G^\circ$ for a reaction with $K = 0.090$ at 25°C.

(a) 6.0 kJ

(b) $-6.0$ kJ

(c) 0.50 kJ

(d) 22 kJ

19. Which of the following processes are spontaneous?

(a) Salt dissolves in $H_2O$

(b) Iron rusts

(c) A clear solution becomes uniform color after dye is added

(d) All of the above

20. Calculate $\Delta S^\circ$ for the reaction:

$$2\,H_2S(g) + SO_2(g) \longrightarrow 3\,S_{\text{rhombic}}(s) + 2\,H_2O(g)$$

Given $S^\circ$ values (J/K · mol): $H_2S(g) = 206$, $SO_2(g) = 248$, $S(s) = 32$, $H_2O(g) = 189$.

(a) +186 J/K

(b) $-186$ J/K

(c) $-474$ J/K

(d) +474 J/K

21. In reaction 1, seven particles in solution form one particle. In reaction 2, four particles form one. Which reaction has a more favorable $\Delta S$?

(a) Reaction 1

(b) Reaction 2

(c) Both are equal

(d) Neither is favorable

22. For an endothermic reaction, does the temperature of the calorimeter increase or decrease?

(a) Increase

(b) Decrease

(c) Remain the same

(d) Increase then decrease

23. If the internal energy of a thermodynamic system is increased by 300. J while 75 J of expansion work is done, how much heat was transferred?

(a) 225 J

(b) 375 J

(c) $-375$ J

(d) $-225$ J

24. Which of the following represents an increase in entropy?

(a) freezing of water

(b) boiling of water

(c) crystallization of salt

(d) $2\,NO(g) \longrightarrow N_2O_2(g)$

25. Calculate the entropy change for methanol ($CH_3OH$) going from a liquid to vapor. ($\Delta H_{vap} = 35.3$ kJ/mol, $T_b = 64.2°C$).

(a) 600 J/K · mol

(b) 551 J/K · mol

(c) 105 J/K · mol

(d) −105 J/K · mol

26. Any irreversible process results in an overall increase in:

(a) Entropy

(b) Enthalpy

(c) Free Energy

(d) Temperature

27. $\Delta S°$ is positive for the reaction:

(a) $Pb(NO_3)_2(aq) + 2\,KI(aq) \longrightarrow PbI_2(s) + 2\,KNO_3(aq)$

(b) $2\,H_2O(g) \longrightarrow 2\,H_2(g) + O_2(g)$

(c) $H_2O(g) \longrightarrow H_2O(s)$

(d) $NO(g) + O_2(g) \longrightarrow NO_2(g)$

28. For a reaction to be spontaneous under standard conditions at all temperatures, the signs of $\Delta H°$ and $\Delta S°$ must be:

(a) $+, +$

(b) $+, -$

(c) $-, +$

(d) $-, -$

29. The equilibrium constant for a reaction is 0.48 at 25°C. What is the value of $\Delta G°$ (kJ/mol)?

(a) 1.8 kJ/mol

(b) −4.2 kJ/mol

(c) $1.5 \times 10^2$ kJ/mol

(d) 4.2 kJ/mol

30. For any reaction at equilibrium, which of the following is true?

(a) $\Delta H < 0$

(b) $\Delta S = 0$

(c) $\Delta H = 0$

(d) $\Delta G = 0$

31. Find the temperature (in K) above which a reaction with a $\Delta H°$ of 53.0 kJ/mol and a $\Delta S°$ of 100.00 J/K · mol becomes spontaneous.

(a) 530 K

(b) 0.53 K

(c) 100 K

(d) 273 K

32. If a process is exothermic and not spontaneous, then what must be true?

(a) $\Delta S > 0$

(b) $\Delta H > 0$

(c) $\Delta G = 0$

(d) $\Delta S < 0$

33. When ice melts at its normal melting point (273.16 K, 1 atm), which of the following is true?

(a) $\Delta H < 0, \Delta S > 0$

(b) $\Delta H > 0, \Delta S < 0$

(c) $\Delta H = 0, \Delta S = 0$

(d) $\Delta H > 0, \Delta S > 0$

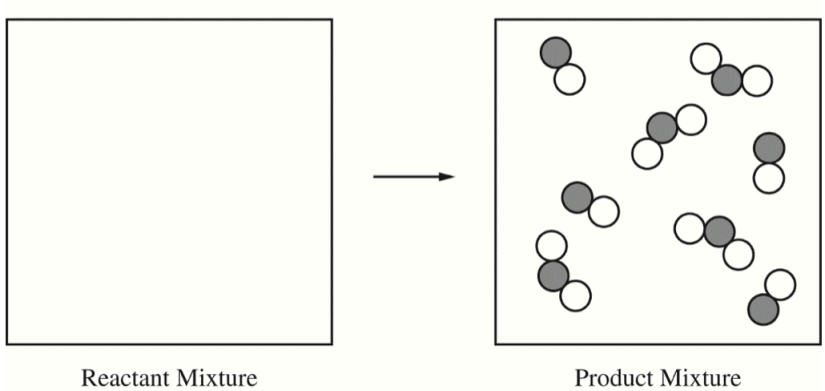
**Free Response Questions (FRQ)** *2 pts each*

1. A student investigates the reactions of nitrogen oxides. One of the reactions in the investigation requires an equimolar mixture of $NO(g)$ and $NO_2(g)$, which the student produces by using the reaction represented below.

$$2\,NO(g) + O_2(g) \longrightarrow 2\,NO_2(g)$$

(a) The particle-level representation of the equimolar mixture of $NO(g)$ and $NO_2(g)$ in the flask at the completion of the reaction between $NO(g)$ and $O_2(g)$ is shown below in the box on the right. In the box below on the left, draw the particle-level representation of the reactant mixture of $NO(g)$ and $O_2(g)$ that would yield the product mixture shown in the box on the right. In your drawing, represent oxygen atoms and nitrogen atoms as indicated below.

| Oxygen atom = ○ | Nitrogen atom = • |
|---|---|




The student reads in a reference text that $NO(g)$ and $NO_2(g)$ will react as represented by the equation below. Thermodynamic data for the reaction are given in the table below the equation.

$$NO(g) + NO_2(g) \rightleftharpoons N_2O_3(g)$$

| $\Delta H_{298}$ | $\Delta S_{298}$ | $\Delta G_{298}$ |
|---|---|---|
| $-40.4$ kJ/mol$_{rxn}$ | $-138.5$ J/(K $\cdot$ mol$_{rxn}$) | 0.87 kJ/mol$_{rxn}$ |

(b) The student begins with an equimolar mixture of $NO(g)$ and $NO_2(g)$ in a rigid reaction vessel and the mixture reaches equilibrium at 298 K.

i. Calculate the value of the equilibrium constant, $K$, for the reaction at 298 K.

ii. If both $P_{NO}$ and $P_{NO_2}$ in the vessel are initially 1.0 atm, will $P_{N_2O_3}$ at equilibrium be equal to 1.0 atm? Justify your answer.

(c) The student hypothesizes that increasing the temperature will increase the amount of $N_2O_3(g)$ in the equilibrium mixture. Indicate whether you agree or disagree with the hypothesis. Justify your answer.

2. A student performs a calorimetry experiment to determine the enthalpy of reaction for the dissolution of calcium carbonate in hydrochloric acid.

$$CaCO_3(s) + 2\,HCl(aq) \longrightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$$

In order to measure the enthalpy of the reaction shown, the student repeats trial 1 by mixing 50.0 mL of HCl(aq) with 1.00 g of $CaCO_3$(s) using a coffee cup calorimeter. The student records the temperature of the system every 20 seconds. The data are given in the following table.

| Time (s) | Measured Temperature of Solution (°C) |
|---|---|
| 0 | 21.20 |
| 20 | 21.51 |
| 40 | 21.70 |
| 60 | 21.85 |
| 80 | 21.90 |
| 100 | 21.90 |

(a) Is the reaction endothermic or exothermic? Justify your answer using the information in the table.

(b) Based on the experimental data, the mass of the system is 51.0 g, and the specific heat of the reaction mixture is 4.0 J/(g $\cdot$° C).

i. Calculate the magnitude of heat transfer, $q$, in joules.

ii. Calculate the enthalpy of reaction in units of kJ/$mol_{rxn}$. Include the algebraic sign on your answer.

# Answer Key

## Multiple Choice Answers:

1. **A** (Plot a). In an exothermic reaction, the products are lower in energy than the reactants. *[Zumdahl Section 6, Page 265, Question 2]*

2. **B** (35 kJ). Use $\Delta E = q + w$. Since $q = +82$ kJ (absorbs) and $w = -47$ kJ (does work), $\Delta E = 82 - 47 = 35$ kJ. *[Zumdahl Section 6, Page 266, Question 21b]*

3. **D** (+77 J). Use $\Delta E = q + w$. Here $w = +100$ J (work on system) and $q = -23$ J (heat released), so $\Delta E = 100 - 23 = 77$ J. *[Zumdahl Section 6, Page 267, Question 24a]*

4. **C** (286 kJ). The $\Delta H$ is for 2 moles of $H_2O$. For 1 mole, it is $572/2 = 286$ kJ. *[Zumdahl Section 6, Page 267, Question 36a]*

5. **A** (900 J). Use $q = mC\Delta T$. $q = (150.0\text{ g})(0.24\text{ J/g}\cdot^\circ\text{C})(298 - 273) = 900$ J. *[Zumdahl Section 6, Page 268, Question 42a]*

6. **A** (−66 kJ/mol). $q = mC\Delta T = 100 \cdot 4.18 \cdot 0.8 = 334.4$ J. Moles $= 0.050\text{ L} \cdot 0.100\text{ M} = 0.005$ mol. $\Delta H = -0.3344\text{ kJ}/0.005\text{ mol} = -66.8$ kJ/mol. *[Zumdahl Section 6, Page 268, Question 51]*

7. **B** (−220.8 kJ). Multiply the first equation by 2 and subtract 2 times the second: $2(-393.7) - 2(-283.3) = -220.8$ kJ. *[Zumdahl Section 6, Page 268, Question 57]*

8. **A** ($\text{Na(s)} + \frac{1}{2}\text{Cl}_2\text{(g)} \longrightarrow \text{NaCl(s)}$). Enthalpy of formation requires elements in their standard states forming 1 mole of product. *[Zumdahl Section 6, Page 269, Question 65]*

9. **B** (Work done by the system to push back the atmosphere). The difference between $\Delta H$ ($q_p$) and $\Delta E$ is $P\Delta V$, which is the work done against the atmosphere. *[Zumdahl Section 6, Page 268, Question 39]*

10. **B** (−55.7 kJ). Molar mass of $CH_4 \approx 16$ g/mol. Heat $= (-891\text{ kJ/mol}) \times (1.00\text{ g})/(16\text{ g/mol}) = -55.7$ kJ. *[Zumdahl Section 6, Page 267, Question 38a]*

11. **B** (Negative). $\Delta S_{surr} = -\Delta H/T$. Since vaporization is endothermic ($+\Delta H$), $\Delta S_{surr}$ is negative. *[Zumdahl Section 16, Page 783, Question 25a]*

12. **B** (7450 J/K). $\Delta S_{surr} = -\Delta H/T = -(-2221000\text{ J})/298\text{ K} = 7453$ J/K. *[Zumdahl Section 16, Page 784, Question 26a]*

13. **B** (630 K). At boiling point, $\Delta G = 0$, so $T = \Delta H/\Delta S = 58510/92.92 = 629.7$ K. *[Zumdahl Section 16, Page 784, Question 30]*

14. **B** (Negative). The reaction goes from 3 moles of gas to liquid; entropy decreases significantly. *[Zumdahl Section 16, Page 784, Question 33c]*

15. **B** ($C_2H_5OH$(g)). Gases have significantly higher entropy than liquids. *[Zumdahl Section 16, Page 784, Question 35b]*

16. **A** (−5.40 kJ). $\Delta G = \Delta H - T\Delta S = -58.03 - (298 \times -0.1766) = -5.40$ kJ. *[Zumdahl Section 16, Page 785, Question 47]*

17. **B** (328.6 K). $T = \Delta H/\Delta S = -58.03/-0.1766 = 328.6$ K. *[Zumdahl Section 16, Page 785, Question 47]*

18. **A** (6.0 kJ). $\Delta G^\circ = -RT\ln(K) = -(8.314 \times 10^{-3} \times 298 \times \ln(0.090)) \approx 5.96$ kJ. *[Zumdahl Section 16, Page 787, Question 74a]*

19. **D** (All of the above). These are classic examples of spontaneous processes. *[Zumdahl Section 16, Page 783, Question 19]*

20. **B** ($-186$ J/K). $\Delta S = \Sigma S_{prod} - \Sigma S_{react} = (3\times32+2\times189)-(2\times206+248) = 474-660 = -186$ J/K. *[Zumdahl Section 16, Page 784, Question 37a]*

21. **B** (Reaction 2). A smaller decrease in disorder (fewer particles lost) results in a more favorable entropy. *[Zumdahl Section 16, Page 786, Question 81]*

22. **B** (Decrease). The system absorbs heat from the calorimeter, causing the calorimeter's temperature to drop. *[Zumdahl Section 6, Page 265, Question 7]*

23. **B** (375 J). $\Delta E = q + w \rightarrow 300 = q - 75 \rightarrow q = 375$ J. *[Zumdahl Section 6, Page 267, Question 23]*

24. **B** (Boiling). Vaporization increases randomness/microstates. *[New Jersey Center for Teaching and Learning, Page 1, Question 1]*

25. **C** (105 J/K · mol). $\Delta S = \Delta H/T = 35300$ J/(64.2 + 273.15) K $\approx 104.6$ J/K · mol. *[New Jersey Center for Teaching and Learning, Page 1, Question 2]*

26. **A** (Entropy). Defined by the Second Law of Thermodynamics. *[New Jersey Center for Teaching and Learning, Page 2, Question 7]*

27. **B** ($2\,H_2O(g) \longrightarrow 2\,H_2(g) + O_2(g)$). Increasing the number of moles of gas increases entropy. *[New Jersey Center for Teaching and Learning, Page 3, Question 11]*

28. **C** ($-, +$). Exothermic and increasing entropy is spontaneous at all temperatures. *[New Jersey Center for Teaching and Learning, Page 11, Question 38]*

29. **A** (1.8 kJ/mol). $\Delta G^\circ = -RT\ln(K) = -8.314 \times 298 \times \ln(0.48) \approx 1818$ J $= 1.8$ kJ. *[New Jersey Center for Teaching and Learning, Page 9, Question 31]*

30. **D** (0). By definition of equilibrium. *[New Jersey Center for Teaching and Learning, Page 15, Question 57]*

31. **A** (530 K). $T = \Delta H/\Delta S = 53000/100 = 530$ K. *[New Jersey Center for Teaching and Learning, Page 13, Question 47]*

32. **D** ($\Delta S < 0$). For an exothermic reaction ($-\Delta H$) to be non-spontaneous ($+\Delta G$), $\Delta S$ must be negative. *[New Jersey Center for Teaching and Learning, Page 14, Question 56]*

33. **D** ($\Delta H > 0, \Delta S > 0$). Melting is endothermic and increases disorder. *[New Jersey Center for Teaching and Learning, Page 17, Question 8]*

## Free Response Answers

**Question 1 (Nitrogen Oxides):** *[2018 AP Chemistry FRQ, Question 2]*

(a) **(2 points total)**

- **1 point:** Correctly representing molecules of NO and $O_2$ in the reactant mixture
- **1 point:** Correctly representing atom conservation (8 molecules of NO and 2 molecules of $O_2$)

(b) (i) **(1 point)** Calculate equilibrium constant:

$K = e^{-\Delta G^\circ/RT} = e^{-\frac{870\ \text{J/mol}}{(8.314\ \text{J/mol}\cdot\text{K})(298\ \text{K})}} = e^{-0.351} = 0.70$

(ii) **(1 point)** No, the pressure will not equal 1.0 atm.

$P_{N_2O_3}$ would only equal 1.0 atm if the reaction goes to completion.

**OR**

The value of $K$ indicates that a substantial amount of reactants will be present at equilibrium.

(c) **(1 point)** Disagree.

Because the reaction is exothermic, increasing the temperature of the reaction will favor the formation of the reactants (according to Le Châtelier's principle).

**Question 2 (Calorimetry):** *[2023 AP Chemistry FRQ, Question 3]*

(a) **(1 point)** Exothermic. The solution temperature increases as the reaction proceeds.

(b) (i) **(1 point)** Calculate heat transfer:

$q_{surr} = mc\Delta T = (51.0\ \text{g})(4.0\ \text{J/g}\cdot^\circ\text{C})(21.90^\circ\text{C} - 21.20^\circ\text{C}) = 140\ \text{J}$

(ii) **(1 point)** Calculate enthalpy of reaction:

$q_{sys} = -q_{surr} = -140\ \text{J} = -0.14\ \text{kJ}$

$1.00\ \text{g CaCO}_3 \times \frac{1\ \text{mol CaCO}_3}{100.09\ \text{g CaCO}_3} \times \frac{1\ \text{mol}_{rxn}}{1\ \text{mol CaCO}_3} = 0.00999\ \text{mol}_{rxn}$

$\Delta H_{rxn} = \frac{-0.14\ \text{kJ}}{0.00999\ \text{mol}_{rxn}} = -14\ \text{kJ/mol}_{rxn}$